

A three-cell multi-tile generating a critical Gabor Riesz basis

Candidate full counterexample for Problems 6.1 and 6.2 of arXiv:2202.06343

Autonomous proof packet — agent_lane_18 (GPT5.6)

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Abstract

Let

$$\Omega = [0, 2) \cup [3, 4), \quad \Lambda = \Gamma = \mathbb{Z}.$$

Then Ω is a level-three multi-tile, but not a tile, with respect to $\mathbb{Z}$. Nevertheless, the critical Gabor system $\mathcal{G}(\chi_\Omega, \mathbb{Z} \times \mathbb{Z})$ is a Riesz basis for $L^2(\mathbb{R})$. On the Zak unit square its multiplier is $1 + z + z^3$, and

$$\frac{47 - 14\sqrt{7}}{27} \leq |1 + z + z^3|^2 \leq 9 \quad (|z| = 1).$$

This gives a positive answer to Problem 6.2, even with Γ a lattice, and a negative answer to the intended tiling-necessity question in Problem 6.1. It also contradicts Theorems 1.3 and 1.4 of the source as stated. The source proof incorrectly promotes an almost-everywhere multi-tiling property to continuity of the Zak transform across lattice-cell boundaries.

1 The questions in the source

Frederick and Mayeli ask the following on page 20 of arXiv:2202.06343v1. The image is included to fix the version and wording.

6 Open problems

Problem 6.1. Our characterization in Theorem 1.2 shows that the statement that $\mathcal{G}(\chi_\Omega, \Lambda \times \Gamma)$ forms a Riesz Gabor basis for any time-frequency shift domain $\Lambda \times \Gamma$ is equivalent to the statement that $\mathcal{E}(\Gamma)$ forms a Riesz basis for $L^2(\Omega)$, provided that Ω tiles $\mathbb{R}^d$ by Λ . This result holds under a prior assumption on the structure of Ω . We also observed in Theorem 1.3 that for certain lattices, if Ω multi-tiles $\mathbb{R}^d$ by Λ , for $\mathcal{G}(\chi_\Omega, \Lambda \times \Gamma)$ to form a Gabor Riesz basis, it is necessary that Ω tiles $\mathbb{R}^d$ by Λ . Can we improve these results and find a necessary tiling condition for any separable Riesz time-frequency domain $\Lambda \times \Gamma$ without any prior geometric assumptions on the set Ω ?

The study of the following problem has been motivated by our observation in Example 1.1.

Problem 6.2. Let Ω be a multi-tiling set with respect to a lattice Λ at level $k > 1$. Is it possible to find a countable set Γ (not necessarily a lattice) such that $\mathcal{G}(\chi_\Omega, \Lambda \times \Gamma)$ is a Riesz Basis for $L^2(\mathbb{R}^d)$?

The exact relevant text is:

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The capitalization and punctuation above follow the source. Problem 6.3, concerning a ball, is not addressed here.

2 Definitions and proof intuition

For $g \in L^2(\mathbb{R})$ and $m, n \in \mathbb{Z}$, set

$$M_m g(t) = e^{2\pi i m t} g(t), \quad T_n g(t) = g(t - n).$$

The critical integer Gabor system is $\mathcal{G}(g, \mathbb{Z} \times \mathbb{Z}) = \{M_m T_n g : m, n \in \mathbb{Z}\}$. A sequence (f_j) is a Riesz basis if it is the image of an orthonormal basis under a boundedly invertible operator. Equivalently, it is complete and there are $0 < A \leq B < \infty$ such that

$$A \sum_j |c_j|^2 \leq \left\| \sum_j c_j f_j \right\|_2^2 \leq B \sum_j |c_j|^2$$

for every finitely supported scalar family (c_j) .

The Zak transform, with the convention used in the source, is

$$Zf(x, \xi) = \sum_{k \in \mathbb{Z}} f(x + k) e^{2\pi i k \xi}, \quad (x, \xi) \in [0, 1)^2.$$

It extends to a unitary map from $L^2(\mathbb{R})$ onto $L^2([0, 1)^2)$. Moreover,

$$Z(M_m T_n f)(x, \xi) = e^{2\pi i (mx + n\xi)} Zf(x, \xi). \quad (1)$$

Intuition. If Ω is made from finitely many integer translates of $[0, 1)$, then $Z\chi_\Omega$ is a finite 0/1 Laurent polynomial in $z = e^{2\pi i \xi}$. At critical density, equation (1) says that the Gabor system is exactly the standard Fourier basis of the Zak square multiplied by this polynomial. It is therefore a Riesz basis if and only if the polynomial never vanishes on the unit circle. Two-term masks always vanish, but the three-term mask $1 + z + z^3$ does not.

3 A finite-mask criterion

For a nonempty finite set $A \subset \mathbb{Z}$, define, modulo null endpoints,

$$\Omega_A = \bigcup_{a \in A} ([0, 1) + a), \quad p_A(z) = \sum_{a \in A} z^a.$$

Proposition 1 (finite lattice-fiber masks). *The set Ω_A multi-tiles $\mathbb{R}$ by $\mathbb{Z}$ at level $|A|$. Furthermore,*

$$\mathcal{G}(\chi_{\Omega_A}, \mathbb{Z} \times \mathbb{Z}) \text{ is a Riesz basis for } L^2(\mathbb{R}) \iff p_A(z) \neq 0 \text{ for every } z \in \mathbb{T}.$$

When these conditions hold, the optimal squared Riesz bounds are

$$A_0 = \min_{|z|=1} |p_A(z)|^2, \quad B_0 = \max_{|z|=1} |p_A(z)|^2.$$

Proof. Fix a noninteger t and write $t = x + j$ with $0 < x < 1$ and $j \in \mathbb{Z}$. For each $a \in A$, precisely the translate $\Omega_A + (j - a)$ contains t . These $|A|$ translates are distinct, and no other integer translate contains t . Hence the multi-tiling level is $|A|$ almost everywhere.

For $0 \leq x < 1$, again away from irrelevant endpoints,

$$Z\chi_{\Omega_A}(x, \xi) = \sum_{k \in \mathbb{Z}} \chi_{\Omega_A}(x + k) e^{2\pi i k \xi} = \sum_{a \in A} e^{2\pi i a \xi} = p_A(e^{2\pi i \xi}).$$

Let $E_{m,n}(x, \xi) = e^{2\pi i(mx + n\xi)}$. The family $(E_{m,n})_{m,n \in \mathbb{Z}}$ is an orthonormal basis of $L^2([0, 1]^2)$. By (1), the Zak transform carries the proposed Gabor system to $(p_A(e^{2\pi i \xi})E_{m,n})_{m,n \in \mathbb{Z}}$.

If p_A is zero-free on $\mathbb{T}$, compactness gives $0 < A_0 \leq B_0 < \infty$. Thus multiplication by $p_A(e^{2\pi i \xi})$ is boundedly invertible on the Zak square. Its image of the Fourier orthonormal basis is a Riesz basis, with the displayed bounds. Conversely, if p_A has a unit-circle zero, continuity makes its essential infimum zero. Multiplication by it is not bounded below, so its image of the Fourier basis cannot be a Riesz basis. $\square$

4 The counterexample

Theorem 1. *Let*

$$\Omega = [0, 2) \cup [3, 4), \quad \Lambda = \Gamma = \mathbb{Z}, \quad g = \chi_{\Omega}.$$

Then:

1. Ω multi-tiles $\mathbb{R}$ by Λ at level three and does not tile $\mathbb{R}$ by Λ at level one;
2. $\mathcal{G}(g, \Lambda \times \Gamma)$ is a Riesz basis for $L^2(\mathbb{R})$;
3. its optimal squared Riesz bounds are

$$\frac{47 - 14\sqrt{7}}{27} \quad \text{and} \quad 9.$$

Consequently Problem 6.2 has a positive answer, even with Γ a lattice, and tiling by Λ is not necessary in Problem 6.1.

Proof. Here $\Omega = \Omega_A$ for $A = \{0, 1, 3\}$, so Proposition 1 gives the level-three assertion and reduces the Riesz-basis claim to the polynomial

$$p(z) = 1 + z + z^3.$$

Suppose $z = e^{i\theta}$ and $p(z) = 0$. Taking imaginary parts yields

$$0 = \sin \theta + \sin 3\theta = 2 \sin 2\theta \cos \theta = 4 \sin \theta \cos^2 \theta.$$

Thus θ is congruent modulo 2π to one of $0, \pi, \pi/2, 3\pi/2$. Direct substitution gives, respectively, $p(z) = 3, -1, 1, 1$, a contradiction. Hence p is zero-free on $\mathbb{T}$, proving the Riesz-basis assertion.

For the exact bounds, put $c = \cos \theta$. A direct expansion gives

$$\begin{aligned} |1 + e^{i\theta} + e^{3i\theta}|^2 &= 3 + 2(\cos \theta + \cos 2\theta + \cos 3\theta) \\ &= 8c^3 + 4c^2 - 4c + 1 =: F(c). \end{aligned}$$

On $[-1, 1]$, the only interior critical points are

$$c_{\pm} = \frac{-1 \pm \sqrt{7}}{6}.$$

Using $6c_{\pm}^2 + 2c_{\pm} - 1 = 0$ gives

$$F(c_+) = \frac{47 - 14\sqrt{7}}{27}, \quad F(c_-) = \frac{47 + 14\sqrt{7}}{27}.$$

Together with $F(-1) = 1$ and $F(1) = 9$, this proves the displayed minimum and maximum. Finally, the level-three count rules out level-one tiling by the same lattice $\Lambda = \mathbb{Z}$. $\square$

5 Why the source’s obstruction does not apply

The example is not merely outside Theorem 1.3 of the source: for $M = N = 1$ it meets that theorem’s stated hypotheses and contradicts its conclusion. The same is true of the equivalence asserted in Theorem 1.4.

The standard critical-density statement used by the source is sound: if a Zak transform is continuous on all of $\mathbb{R}^2$ with the usual quasi-periodicity, then it must vanish. But an arbitrary multi-tiling indicator need not have a continuous Zak transform. In the present example, for $0 < x < 1$ its lattice fiber is $\{0, 1, 3\}$, whereas for $1 < x < 2$ the fiber is $\{-1, 0, 2\}$. Thus

$$Zg(x, \xi) = p(e^{2\pi i \xi}) \quad (0 < x < 1),$$

while crossing the boundary $x = 1$ changes the expression by the nontrivial quasi-periodic phase $e^{-2\pi i \xi}$. Except at integer ξ , the two one-sided limits differ.

The source’s Lemma 5.3 argues, after saying *without loss of generality, assume that Ω is open*, that the finite lattice fiber is locally constant at every point. Neither step follows from an almost-everywhere multi-tiling hypothesis. Replacing Ω by the open representative $(0, 2) \cup (3, 4)$ changes no L^2 or tiling assertion, but the fiber still jumps at integer cell boundaries. The claimed continuity in Theorem 5.4 therefore fails, and the zero theorem cannot be invoked.

6 Independent checks, novelty, and limitations

The companion script `verify_counterexample.py` checks the exact critical values symbolically, samples the multi-tiling count away from the null boundary set, and performs a million-point numerical audit of the mask. The proof above is independent of that computation.

The polynomial $1 + z + z^3$ and the same lower bound occur in an unrelated positive-definite Toeplitz quadratic-form example in arXiv:math/0507185. Searches of the local arXiv corpus and the web by the source title, exact problem wording, and mask polynomial found no published correction of arXiv:2202.06343 or use of this mask to answer Problems 6.1–6.2. This is a novelty search, not a proof of novelty.

The packet does not address Problem 6.3 about balls, does not classify all separable Gabor Riesz bases, and does not claim that every level- k multi-tile works. Because the result directly challenges stated theorems in the source preprint, a specialist should verify the Zak convention and then notify the authors before dissemination.

Human-review checklist

Item	Check
Multi-tiling	Confirm the three translates covering each noninteger point.
Zak covariance	Confirm signs in (1); signs do not affect the zero-free conclusion.
Polynomial bound	Recheck the four critical values of $F(c)$.
Source contradiction	Compare hypotheses of Theorems 1.3–1.4 with $M = N = 1$.
Novelty	Search journal versions, errata, MathSciNet, and citing literature.

References

1. C. Frederick and A. Mayeli, *A characterization of Gabor Riesz bases with separable time-frequency shifts*, arXiv:2202.06343v1 (2022).
2. K. Gröchenig, *Foundations of Time-Frequency Analysis*, Birkhäuser, 2001. (Standard Zak-transform characterization at critical density.)
3. X. Huang and L. You, *On the strong positive definiteness of infinite matrices*, arXiv:math/0507185 (2005). (Unrelated occurrence of $1 + z + z^3$ and its lower bound.)